

Sana Ullah

Welding QC Inspector

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Riyadh, KSA



Profile

Graduated Mechanical Engineer from Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI), 2024, with recent exposure as a QC Welding Inspector. CSWIP 3.1 Certified, skilled in piping inspection, welding verification, material inspection, and QA/QC documentation. Experienced in reviewing ITPs, MIRs, WIRs, MARs, shop drawings, method statements, and RFIs, ensuring compliance with ASME, API, ASTM, and client specifications. Seeking to leverage technical expertise and quality assurance skills in a QC Welding/Piping role within reputable manufacturing, contracting, or industrial organizations.

Education

Bachelor of Science in Mechanical Engineering

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Pakistan.

2020-2024

Topi, Pak

Experiences

Designation: QC Welding Inspector

Company: HUAYU Arabian Contracting Company.

Project: PP12 Dhurma Power Plant Expansion KSA.

Client: SEC, KSA.

2026-Present

Dhurma, KSA

- Perform welding inspection activities for carbon steel (CS) and stainless steel (SS) piping systems in accordance with ASME B31.3, ASME Section IX, approved WPS, and project ITP requirements.
- Carry out inspection of structural welding activities (including Fin Fan coolers and Air-Cooled Condensers) in line with project specifications and applicable structural codes such as AWS D1.1.
- Review and verify WPS, PQR, WPQ, and welder qualifications, ensuring validity and compliance prior to commencement of welding for both piping and structural works.
- Conduct visual inspection (VT) at all stages (fit-up, in-process, and final) for piping welds and structural joints, ensuring proper weld profile, alignment, dimensions, and adherence to approved procedures.
- Coordinate, monitor, and witness NDT activities including PT, MT, RT, and UT for piping systems, and VT/MT/UT for structural components, ensuring compliance with project specifications and IPP/SEC requirements.
- Verify material compliance for CS and SS, including review of MTCs, heat number identification, PMI (if required), and full traceability of piping and structural materials at site.
- Monitor PWHT activities for applicable piping welds, ensuring compliance with approved procedures, including temperature control, soaking time, heating and cooling rates, and verification of PWHT charts and records.

- Inspect and ensure proper installation and alignment of piping systems, supports, and structural assemblies such as Fin Fan units and Air-Cooled Condensers.
- Prepare, review, and close all quality documentation including WIRs (Welding Inspection Requests), MIRs (Material Inspection Reports), NCRs (Non-Conformance Reports), and other inspection records in accordance with project quality requirements.
- Coordinate effectively with construction teams, QA/QC department, NDT personnel, and client representatives to resolve welding and quality-related issues and ensure smooth execution of piping and structural works.

Designation: Graduate Trainee Engineer

Sep 2024 – Sep 2025

Company: Descon Engineering

Lahore, Pakistan

- Gained practical exposure to mechanical and piping systems, supporting senior QA/QC engineers in inspection, quality control, and project documentation activities.
- Assisted in reviewing piping isometric drawings, P&IDs, and fabrication drawings, ensuring compliance with project specifications and applicable codes such as ASME B31.3 and ASME Section IX.
- Participated in fit-up and welding inspections, verifying joint preparation, alignment, root gap, and adherence to approved WPS requirements.
- Supported in preparation and updating of weld maps, ensuring proper identification of weld joints, traceability of welders, and linkage with NDT and inspection records.
- Assisted in maintaining and updating welding summary reports, including weld joint status, NDT progress, repair tracking, and final acceptance records.
- Contributed to material inspection and verification, including checking MTCs, heat numbers, and ensuring proper material traceability at site/store.
- Coordinated with NDT personnel for execution of RT, UT, PT, and MT activities, and assisted in reviewing reports for compliance with acceptance criteria.
- Supported in preparation and submission of inspection reports (WIRs, MIRs) and follow-up on NCRs to ensure timely closure of quality issues.
- Assisted in monitoring PWHT activities, including checking temperature records and ensuring compliance with approved procedures.
- Worked closely with construction, QA/QC team, and supervisors to ensure smooth workflow, timely inspections, and adherence to project quality requirements.

Designation: Mechanical Internee

Jul 2023 – Aug 2023

Company: Pakistan Welding Institute

Islamabad, Pakistan

- Gained hands-on training in industrial welding processes and procedures, understanding the fundamentals of weld quality, joint preparation, and welding parameters to achieve defect-free welds.
- Performed practical welding using TIG (GTAW), GMAW (MIG), SMAW, and SAW processes on sample plates, focusing on arc control, bead appearance, penetration, and minimizing welding defects.
- Assisted in evaluating weld quality through visual inspection (VT) and basic defect identification such as porosity, undercut, and lack of fusion, improving understanding of acceptance criteria.
- Learned the importance of WPS (Welding Procedure Specification) and followed standard welding parameters including current, voltage, travel speed, and electrode selection.
- Observed and analyzed different welding techniques to compare efficiency, weld strength, and consistency, contributing to improved workmanship and reduced rework.
- Achieved improved welding results by controlling parameters effectively, contributing to a reduction in material rework and scrap by approximately 15% during training exercises.

Leadership Experience

IMechE GIKI Chapter

Feb 2021 – Apr 2024

Event Coordinator – WRE (5th International Workshop on Functional Reverse Engineering of Machine Tools)

- Led a highly successful international workshop at GIKI in collaboration with the Department of Science and Technology, Government of Khyber Pakhtunkhwa, Pakistan.
- Facilitated knowledge exchange and collaboration through keynote talks, technical sessions, and networking opportunities, fostering 120 new industry connections.
- Contributed to the development of indigenous expertise in functional reverse engineering and promoted enterprise of reverse engineering in Khyber Pakhtunkhwa.

Event Coordinator – AKDC (Abul Kalam Azad Design Competition)

- Managed a competitive design competition at GIKI, fostering innovation and creativity among engineering students.
- Oversaw 800 submissions and awarded prizes to the most promising designs, recognizing the top 10 teams for their outstanding work.

Academic Projects

Maglev Vehicle: Design and Development of Linear Translator for Maglev Vehicle

- Engineered a high-performance permanent magnet SLSM for industrial automation, achieving linear velocity of 3 – 5 m/s and an average force of 56 N, enhancing motion precision and efficiency.
- Obtained simulations, optimizing design parameters to achieve a 20% reduction in cogging force and a decrease in operational noise, significantly improving system smoothness and reducing noise pollution.
- Developed an advanced control system using Arduino Mega and BTS7960 motor drivers, operating within a 12 – 18V range and drawing 3 – 5A current, enabling precise motion and directional control within a 3-phase PWM signal.

Wire Bending Machine

- Design and development of wire bending machine.
- Achieved desired performance by bending a wire of few mm thickness.

Transmission Shaft for Automobile Applications

- Design and simulation of transmission shaft.
- Optimized the transmission shaft using aluminum 1100 – H14, reducing weight by 62.5% while maintaining strength and fuel efficiency.
- Performed several FEA simulations to validate the design, ensuring it handles 70.53 Nm torque within a safety factor of 3 and minimal deflection.

Solar Water Heater

- Design and fabrication of solar water heater.
- Used solar energy as a cheap energy source.
- Reliable and cost-efficient supply of hot water for household operations.

ISO Audit

- Executed thorough internal audit of DYNEA Pakistan Limited for 9001:2015 compliance.
- Conducted detailed review of over 20 documents, completed 15+ onsite inspections, and interviewed 7 key personnel to ensure strict adherence to quality management standards.

Certification

- CSWIP 3.1 – Certified Welding Inspector
- WMS (Work Management System) – ADNOC Approved
- Project Management – Google
- SolidWorks – Udemy

Software Skills

- Solid Works
- Thinker CAD
- MS Office
- Hourly Analysis Program
- Auto CAD